Skills Summary

Composing Piecewise and Nonlinear Functions

Use this reference while completing your worksheet or when studying.

Skill 1 — Writing the composed rule

Definition. $(f \circ g)(x) = f\big(g(x)\big)$: replace *every* x in the rule for f by the whole expression $g(x)$, then simplify.

Order matters: $(f \circ g)$ and $(g \circ f)$ are different functions.

Example: $f(x) = x^2 - 2x$, $g(x) = x + 1$. Write $(f \circ g)(x)$.
Step 1. Both x's in the rule for f must be replaced — substituting into one of them is the standard slip, so do it before any expanding.
Step 2. $(x+1)^2 - 2(x+1) = x^2 + 2x + 1 - 2x - 2$.
$\Rightarrow$ $\boldsymbol{(f \circ g)(x) = x^2 - 1}$

Try it: $f(x) = x^2 + 4x$, $g(x) = x - 2$. Write $(f \circ g)(x)$.

check: $\boldsymbol{(f \circ g)(x) = x^2 - 4}$

Skill 2 — Evaluating through a piecewise rule

Two-step rule. Compute the inner value first. Then compare *that* number — not the original input — with the boundary to pick the branch.

In $(f \circ f)(a)$ the branch is chosen twice, and often differently each time.

Example: $g(x) = 2x - 3$ and $f(x) = x + 6$ for $x < 1$, $f(x) = x^2$ for $x \geq 1$. Find $(f \circ g)(5)$.
Step 1. The branch depends on the value entering f, so compute $g(5)$ before looking at the cases at all.
Step 2. $g(5) = 7$, and $7 \geq 1$, so use x^2: $7^2 = 49$.
$\Rightarrow$ $\boldsymbol{(f \circ g)(5) = 49}$

Try it: Same f and g. Find $(f \circ g)(1)$.

check: $\boldsymbol{(f \circ g)(1) = 5}$

(!) Watch out: testing the original input against the boundary is the most common piecewise-composition error. The boundary belongs to f, so only the number entering f gets tested.

Skill 3 — Solving a composition equation

Method. Write $(f \circ g)(x)$ as one expression, set it equal to the target, and solve. With a square root, isolate it and square both sides — then check every candidate in the *original* equation.

Example: $f(x) = \sqrt{x+1}$, $g(x) = 3x$. Solve $(f \circ g)(x) = 4$.
Step 1. Compose first so there is a single radical to isolate, rather than two functions to juggle.
Step 2. $\sqrt{3x+1} = 4$, so $3x + 1 = 16$ and $3x = 15$; check: $\sqrt{16} = 4$.
$\Rightarrow$ $\boldsymbol{x = 5}$

Try it: $f(x) = \sqrt{x-2}$, $g(x) = 4x$. Solve $(f \circ g)(x) = 3$.

check: $\boldsymbol{x} = \frac{11}{4}$